

ROTTERDAM THE HAGUE AP, Netherlands

WMO: 063440

Lat: 51.961N

Lon: 4.447E

Elev: -5

StdP: 101.38

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.9	(c) -3.9	(d) -9.7	(e) 1.6	(f) -4.5	(g) -7.3	(h) 2.0	(i) -3.0	(j) 14.6	(k) 9.2	(l) 13.3	(m) 9.6	(n) 3.1	(o) 60	(p) 0.581

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	28.4	20.0	26.1	19.2	24.2	18.2	21.0	26.5	20.0	24.6	19.0	23.0	3.9	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	13.9	23.2	18.2	13.1	22.0	17.4	12.4	21.0	60.7	26.6	57.0	24.8	53.9	23.1	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.8	(b) 10.4	(c) 9.2	(e) -8.7	(f) 32.3	(g) 3.0	(h) 1.7	(i) -10.9	(j) 33.6	(k) -12.6	(l) 34.6	(m) -14.3	(n) 35.6	(o) -16.5	(p) 36.8		
(a) 11.8	(b) 10.4	(c) 9.2	(e) -9.1	(f) 23.1	(g) 2.9	(h) 0.9	(i) -11.2	(j) 23.7	(k) -12.9	(l) 24.2	(m) -14.6	(n) 24.7	(o) -16.7	(p) 25.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	3.9	4.3	6.5	9.6	13.0	15.9	18.2	17.9	15.1	11.6	7.6	4.6
	DBStd	6.11	4.00	3.76	3.24	3.52	3.44	2.98	2.89	2.59	2.58	3.21	3.32	4.05
	HDD10.0	802	191	161	114	49	9	0	0	0	0	21	86	172
	HDD18.3	2861	448	393	367	261	168	86	38	39	101	209	323	427
	CDD10.0	1063	1	1	6	38	104	176	253	246	154	70	12	3
	CDD18.3	81	0	0	0	0	4	11	33	27	5	1	0	0
	CDH23.3	603	0	0	0	4	42	103	249	177	27	1	0	0
	CDH26.7	146	0	0	0	0	6	23	69	43	4	0	0	0
Wind	WSAvg	4.3	5.2	5.0	4.8	4.1	4.0	3.8	3.7	3.5	3.7	4.2	4.4	4.9
Precipitation	PrecAvg	901	75	68	52	43	57	71	86	99	85	83	95	87
	PrecMax	1208	171	133	106	106	120	149	189	241	244	169	160	168
	PrecMin	638	8	19	11	0	23	26	15	21	18	14	7	16
	PrecStd	132	38	29	27	26	27	36	44	57	54	38	37	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.5	13.8	17.7	23.7	27.2	29.6	31.7	30.5	26.3	21.9	16.3	13.3
		MCWB	11.0	10.0	12.4	15.3	18.7	20.4	20.9	21.0	19.5	17.2	14.0	11.9
	2%	DB	11.1	11.6	15.0	20.5	24.2	26.3	28.6	27.7	23.3	19.0	14.6	12.3
		MCWB	9.9	9.4	10.9	14.0	16.5	19.1	20.2	20.0	18.3	16.0	13.0	11.1
	5%	DB	10.0	10.5	12.8	17.9	21.8	23.8	26.0	25.1	21.2	17.3	13.4	11.2
		MCWB	8.8	8.8	9.9	12.6	15.5	17.9	19.4	19.1	17.2	15.1	11.9	10.1
	10%	DB	9.0	9.4	11.3	15.5	19.5	21.5	23.8	23.0	19.5	16.2	12.2	10.0
		MCWB	7.8	8.0	9.1	11.6	14.6	16.7	18.4	18.2	16.2	14.2	10.9	8.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	10.9	13.0	16.6	19.6	21.6	22.4	22.2	20.3	17.7	14.5	12.3
		MCDB	12.1	12.6	16.3	22.3	25.7	27.5	28.2	28.0	24.8	20.8	15.6	13.0
	2%	WB	10.1	9.9	11.5	14.4	17.6	19.7	21.0	20.9	18.9	16.4	13.3	11.2
		MCDB	10.9	11.0	14.0	19.2	22.2	24.7	26.7	26.0	22.4	18.5	14.5	12.1
	5%	WB	8.9	9.0	10.3	13.1	16.1	18.4	20.0	19.7	17.8	15.5	12.1	10.2
		MCDB	9.8	10.2	12.4	17.1	20.5	22.7	24.8	23.8	20.4	17.2	13.3	11.1
	10%	WB	8.0	8.1	9.4	11.9	14.9	17.2	18.9	18.7	16.8	14.5	11.1	9.0
		MCDB	8.9	9.3	11.1	15.0	19.0	20.7	23.0	22.1	19.0	15.9	12.1	9.9
Mean Daily Temperature Range	5% DB	MDBR	4.5	5.3	6.8	8.4	8.7	8.5	8.2	8.2	7.5	6.5	5.1	4.5
		MCDBR	4.5	5.9	9.5	12.1	12.8	12.7	12.3	12.0	9.8	7.4	5.2	4.1
		MCWBR	4.4	4.6	6.3	6.9	6.6	6.6	5.7	5.7	5.5	5.0	4.4	4.0
	5% WB	MCDBR	4.4	5.2	8.2	10.8	11.1	10.9	10.8	10.5	8.7	7.0	5.1	4.1
		MCWBR	4.4	4.4	5.9	6.5	6.4	6.2	5.6	5.6	5.3	5.1	4.6	4.1
Clear-Sky Solar Irradiance	taub	0.342	0.363	0.397	0.423	0.430	0.425	0.433	0.424	0.401	0.377	0.354	0.338	
	taud	2.316	2.302	2.222	2.190	2.228	2.262	2.248	2.289	2.342	2.364	2.334	2.306	
	Ebn at Noon	642	740	784	811	824	832	816	803	776	715	620	576	
	Edh at Noon	65	87	113	131	132	129	129	117	100	81	64	57	
All-Sky Solar Radiation	RadAvg	0.71	1.40	2.60	4.21	5.09	5.46	5.15	4.35	3.16	1.82	0.83	0.54	
	RadStd	0.08	0.26	0.33	0.42	0.36	0.39	0.57	0.36	0.27	0.15	0.08	0.06	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (83 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page